Biologically Inspired Physical Model of the Vocal Tract for the Tasks of Recognition of the Current Psycho-Emotional State of a Person

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Abstract. The urgency of the development and use of a physical biologically inspired model of the human vocal tract (VT) of a person for solving a complex of tasks on recognizing the level of stress based on the processing of speech information is shown. As an advantage of such a biologically inspired model, compared with other types of existing VT models, the possibility of studying the complex effects of nervous tension on speech parameters was highlighted. The volatility of the fundamental frequency (FF) is highlighted as such an informative parameter. The structure of the VT physical model is considered, as well as an example of its practical application. It is shown that improving the reliability of the so-called speech signature technology (SS), which is used to protect important paper documents from counterfeit and falsification, is one of the promising areas of application of the developed model.

Keywords: Biologically inspired model · Stress recognition · Speech signature

1 Introduction

Information technologies based on speech information processing are widely used nowadays in almost all spheres of human activity. These technologies are of particular importance in the field of information security. Thus, the SS technology actively developed recently is an additional level of protection of important documents from their fakes and falsification [1].

Such a signature allows you to increase the level of security, first of all, documents presented in paper form. The SS technology assumes the inclusion of significant fragments of the speech of its author, represented graphically using sonograms, in the document. Reproduction and analysis of speech fragments by the recipient of the document makes it possible to identify his personality and the authenticity of the main provisions of the document. This technology of protection of paper documents does not limit the ability to copy, replicate and scan [1].

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SS technology also makes it possible to ensure that the author of the document, when compiled, was in a normal sane state and did not sign it under duress. To do this, in the SS, it is necessary to preserve and transmit information about the voice features that most characterize a person's current psycho-emotional state. In this regard, it should be noted that in terms of diagnosing the psycho-emotional state, voice information takes the second place in informational content after video information about a person's face.

The conducted research identifies a number of informative features of speech in its psycho-semantic, paralinguistic, psycholinguistic, pragmatic and syntactic analysis [2]. Taking into account, as a rule, the time-limited duration of the SS, the paralinguistic analysis of speech is most appropriate. Such an analysis usually includes an assessment of the rate of speech, its loudness, melody, the dynamics of changes in timbre, jitter of FF. It is possible to determine the gender characteristics of the author of the document. Jitter (variability) of FF is widely used in practice, including for purely medical purposes, to identify a number of serious diseases [3].

From the point of view of human physiology, the so-called tremor of the voice, which is manifested, above all, in the FF tremor, is caused by involuntary muscle tension during stress. First of all, these are the muscles of the vocal cords, the muscles of the neck and the face. The influence of the state of these muscles on the level of tremor of the voice is quite difficult and ambiguous. The individual features of body shaping are one of the factors that aggravate the study of the mechanisms of tremor.

In practice, various VT models are used, allowing simulation to be carried out with some degree of approximation of the mechanisms by which voice tremor occurs and its conditional state of stress. The main problem with the use of such models is the need for verification, which implies a rather large amount of experimental research involving the necessary contingent of people.

The development of a model of human VT, which allows for a complex of studies aimed at identifying patterns of voice trembling at various levels of stress, is an urgent task at present. The application of such a model in practice will reduce the time and material costs in the development of algorithms and tools for reliable determination of the level of stress. This model is of particular importance in the implementation of SS technology with the support of the function of assessing the psycho-emotional state of the author of the document.

The aim of the work is to develop a biologically determined physical model of human VT, focused on the study of the mechanisms of tremor of the gloss.

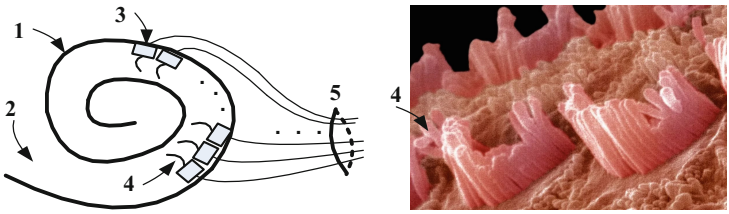
2 Biological Inspiration and the Essence of the Proposed Approach

The approach developed is based on the following biologically inspired principles:

- the hearing aid and the human brain allow you to simultaneously perceive and analyze the spectral composition of acoustic signals in real time;
- the person's nervous tension is manifested both in the tension and trembling of the vocal cords muscles, and of the muscles of the neck and face, which has a complex

effect on FF and, ultimately, manifests itself in characteristic modulations of the entire spectrum of overtones;
– the constitution, age, gender and health conditions (inflammation, swelling of the throat) are the main factors influencing the degree and nature of stress.

In Fig. 1a shows the model of the main recognition part of the human hearing aid. The so-called cochlea of the inner ear 1, having a throat 2, is an acoustic wave guide in which neurons 3 are located with resonating hairs 4.



a) b)

Fig. 1. Model of the human hearing aid.

The parameters of the cochlea waveguide make it possible to perceive oscillations in a wide frequency range of 25 Hz–20 kHz. The auditory nerve 5 transmits information to the appropriate brain center for detailed processing. The huge number of hairs 4 in the cochlea (Fig. 1b [4]) allows simultaneous parallel perception of the parameters of all harmonics of the speech signal. This circumstance is important and is taken into account when developing the human VT model.

From practice, it is known, for example, that at a high level of stress, the voice of a person becomes "metallic". At the same time, in its spectral composition changes, including the energy balance between the individual harmonics, which is clearly perceived by the audience.

The complex nature of the effects of nervous tension must be considered when developing VT models. From this point of view, the main requirements for such models include:

– the ability to simulate the effects of stress and muscle tremor of the vocal cords, neck and face under stress;
– the presence of parameters regulating the degree of such a stress effect;
– the possibility of changing the parameters of the VT model responsible for the individual constitutional characteristics of the person's body;
– focus on the synthesis of vowel sounds, the tremor of which, to the greatest extent, conveys a stressful state;
– ease of implementation, clarity, the possibility of modifying and developing the VT model to describe other effects.
– The complex effect of nervous tension is to some extent taken into account when developing VT models. Among these models should be highlighted:

- mathematical models based on differential equations describing the propagation of acoustic oscillations [5, 6];
- models describing the processes of mechanical oscillations [7];
- physical models [8].

The most complete listed requirements satisfy the physical model. Such models implement the mechanisms of propagation, reflection and resonance of actually existing acoustic oscillations, which makes it possible to model effects that are difficult to give a precise mathematical description.

Studies conducted at the Department of Electrical and Electronic Design at the Japanese University Sophia [8] confirmed the effectiveness of using a physical model that allows synthesizing the main vowel sounds of the Japanese language. The model is an acoustic channel completely identical to the human VT geometry, containing resonant laryngeal cavity, pharynx, mouth, nose, and the cavity between the teeth and lips. The model is tunable. It may contain a different number of fragments with holes of different diameters, which permits to imitate a person's VT 10–20 cm long. For example, to simulate the formation of the Japanese vowel sound "o", you must use 14 fragments with holes of the following diameters: 14 mm, 22 mm, 26 mm, 34 mm, 34 mm, 38 mm, 34 mm, 28 mm, 22 mm, 16 mm, 14 mm, 16 mm, 20 mm, 25 mm.

The disadvantages of the physical model include the limited functional possibilities for studying the effects of the modulation of vowel sounds caused by the tremor of the muscles involved in the speech-forming process under stress. The main factor limiting the functionality is the rigid construction of the model fragments.

The developed biologically inspired physical model of VT is adaptive. Its fragments are made of soft, easily deformable material (silicone). Moreover, each fragment has a built-in mechanism for its deformation (compression) with computer control. This makes it possible to simulate the static and dynamic effects of the influence of muscle activity on the main components of VT. The developed VT physical model allows one to satisfy all the requirements listed above in order to take into account the complex nature of the effects of nervous tension on the gloss tremor.

3 Practical Use of the Developed Biologically Inspired Physical VT Model

In Fig. 2 shows an example of the practical use of the developed model as part of an experimental setup for the study of acoustic manifestations of a person's stress state.

In the example presented, the model is used to synthesize the vowel sound "a". For this, 19 fragments with characteristic window sizes were used. Filling the fragments with plastic material makes it possible to modulate the size of windows under the influence of forces created by actuators F. Taking into account the low-frequency range of action of forces imitating muscular effects on the VT under stressful conditions of varying degrees, electromagnetic performing devices were used.

Controlling actuators using an interface (I) connected to a personal computer (PC) allows you to simulate the tremor of various muscles involved in the process of speech formation.

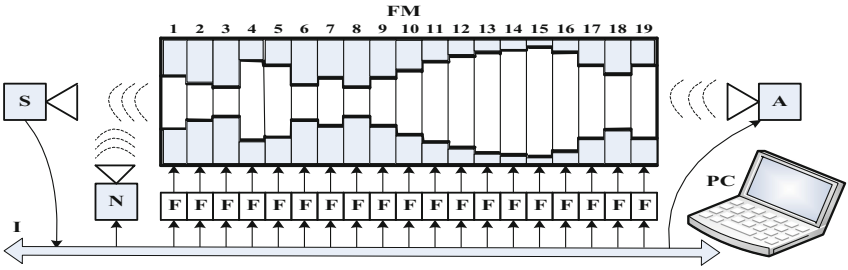


Fig. 2. An example of the use of the developed biologically inspired model.

In order to study the effect of external acoustic noise on the recognition of a tremor in a person's voice, an additional acoustic emitter (N) was used in the experimental setup. This emitter allows you to mix in the signal generated by the FM physical model when it is activated by periodic signals from the source A, the signal interference of a given spectrum and power level. The resulting acoustic signal is recorded by receiver S.

Practical use of the developed model allows solving the following tasks:

- study of the influence of the peculiarities of the constitutional structure of the human body on the information content of various overtones under stress of various degrees;
- the study of the influence of the level and spectral composition of external noise on the accuracy of the determination of FF and its instability;
- study of the degree of influence of the tremor of various muscles that affect the work of VT,
- study of the informativeness of various vowel sounds from the point of view of the reliability of recognition of stress at various levels;
- justification of the optimal set of overtones of vowels, the most informative in terms of recognizing the current level of stress.

4 Experimental Approbation of the Approach

The developed physical model was used to determine the most informative set of overtones that provide reliable recognition of the stress level in the presence of external noise of varying intensity for two vowel sounds of the Russian language.

In Fig. 3 shows examples of sonograms synthesized using the developed model in the presence of a noise component (gray background between brighter lines corresponding to overtones).

It is clearly seen that for the first vowel sound (Fig. 3a), the overtones F14, F19-F21 should be the most informative. Although their intensity loses to the lower-frequency components of F0-F2, the manifestation of tremor is much more pronounced. Similarly, for the second vowel sound (Fig. 3b), the most informative overtones are F14 and F15. The presented examples correspond to the modeling of the complex effect of muscle tremor under stress.

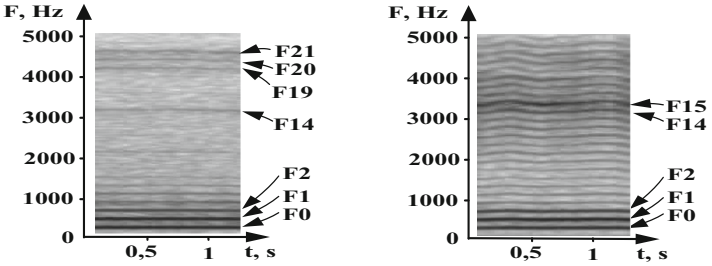


Fig. 3. Examples of sonograms of vowel sounds, obtained using the developed model.

5 Conclusion

The developed VT physical model is biologically inspired and is intended for use in research and training purposes.

The model makes it possible to investigate the effects of a complex effect of a stressful state of a person on VT, which makes it possible to identify the most informative group of speech parameters that reliably identify the state and level of stress.

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